Kubernetes:

Kubernetes is containerisation management tool. It is managing our container. It overcomes the docker's draw backs.

Features of Kubernetes:

- Auto scaling - Kubernetes automatically scale up and scale down the resources as per app demand. In auto scaling they are two types
    1. Vertical auto scaling: It will increase the CPU's, memory and storage
    2. Horizontal auto -scaling: it will increase servers as per app demand.
- Auto healing – In Kubernetes it automatically repairs the pods and create pods within a few seconds. To reduce app down time.
- Load balancing – It distribute the users traffic b/w the servers equally
- Platform independent – It run on any infra or platform
- Fault tolerance – it will notify the issue
- Roll back – switch the version easily
- Orchestration – it will manage the entire k8s cluster and resources.

Cluster:

It is a combination of nodes. Node is virtual machine. In a cluster there are 2 types:

1. Master node
2. Worker node

Master node:

- It will manage the entire cluster and resources.
- It is heart of the cluster
- Without master node you can't create any worker node
- Communicates with worker node

In master node there are some components:

1. API:
   It is entry point of master node. Without this you can't go forward. Whenever you run the commands firstly it will go to API server only. It will validate the requests and update clusters.
2. Etcd:
   It stores our entire data. It will store in the form of key-value pair.
3. Controller manager:
   It is also called as Kube-controller manager. It will manage our cluster always in desired state. Before creating any resources, the cluster always in desired state only.
4. Scheduler:
   It will schedule our cluster tasks. It will decide which node should run on.

Worker Node:

Worker node always communicates with master node. With help of master node you can create n number of worker nodes. We have to deploy the application in the worker node.

In worker node there are so many components:

1. Kube let:
   It is an agent on every node. It will communicate API server in master node. It will create and manage pods
2. Kube proxy:
   It will store the entire network of the cluster. It will communicate to pod services. It will maintain IP tables.
3. Pod:
   It is smallest deployable unit where we are going to manage our containers.
4. Container run time:
   It will run the container inside the k8s.